

3D Avatar Generation & Body Reconstruction Research

Report

AI/ML Internship — Day 17

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1. Introduction

The rapid growth of Artificial Intelligence, Computer Vision, and Virtual Try-On technologies has increased the demand for realistic digital human representations. Modern Virtual Try-On systems rely on accurate 3D avatars and body reconstruction techniques to simulate clothing fit, body proportions, and garment behaviour.

Traditional 2D image-based approaches are often limited in capturing body shape and depth information. To overcome these limitations, advanced AI models generate realistic 3D human avatars from one or more images and reconstruct detailed body geometry for fashion, gaming, AR/VR, healthcare, and e-commerce applications.

This research explores modern 3D body reconstruction and avatar generation techniques, including SMPL, PIFuHD, ICON, HumanNeRF, and 3D Gaussian Splatting.

2. Objectives

The objectives of this research are:

- Study modern 3D avatar generation techniques.
 - Understand AI-based body reconstruction methods.
 - Compare different 3D human modelling approaches.
 - Analyse strengths and limitations of each method.
 - Evaluate suitability for Virtual Try-On applications.
 - Identify the most effective technologies for future implementation.
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3. Importance of 3D Avatar Generation

3D avatar generation plays a critical role in creating realistic virtual experiences. Instead of placing garments on a flat image, 3D avatars allow clothing to interact with the user's body shape and posture.

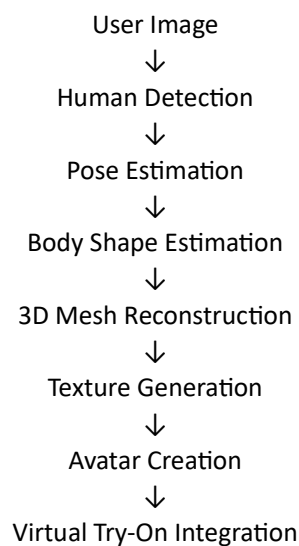
Benefits include:

- Accurate garment fitting

- Personalized user experiences
 - Realistic cloth simulation
 - Virtual shopping experiences
 - Digital fashion applications
 - AR/VR integration
 - Reduced product returns in e-commerce
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4. 3D Human Body Reconstruction Workflow

The general workflow of 3D avatar generation is:



This workflow converts a 2D image into a complete 3D digital representation of the user.

5. SMPL (Skinned Multi-Person Linear Model)

Overview

SMPL is one of the most widely used parametric human body models. It represents the human body using shape and pose parameters and generates realistic 3D human meshes.

Features

- Parametric body representation
- Pose estimation support
- Shape modelling
- Realistic human mesh generation

Advantages

- Industry-standard model
- Fast mesh generation
- Compatible with many AI frameworks
- Strong research support

Limitations

- Limited clothing detail
- Requires additional models for realistic garments

Applications

- Virtual Try-On
- Motion Capture
- Avatar Generation
- Human Body Analysis



Figure 1: SMPL body model

6. PIFuHD (Pixel-Aligned Implicit Function for High-Resolution Human Digitization)

Overview

PIFuHD is an advanced deep-learning framework that reconstructs highly detailed 3D human models from a single image.

Features

- High-resolution reconstruction
- Fine geometric details

- Single-image input
- Detailed clothing reconstruction

Advantages

- High-quality output
- Captures clothing details
- Realistic body reconstruction

Limitations

- Computationally expensive
- Longer processing time

Applications

- Fashion Technology
- Digital Humans
- Virtual Try-On
- Character Creation

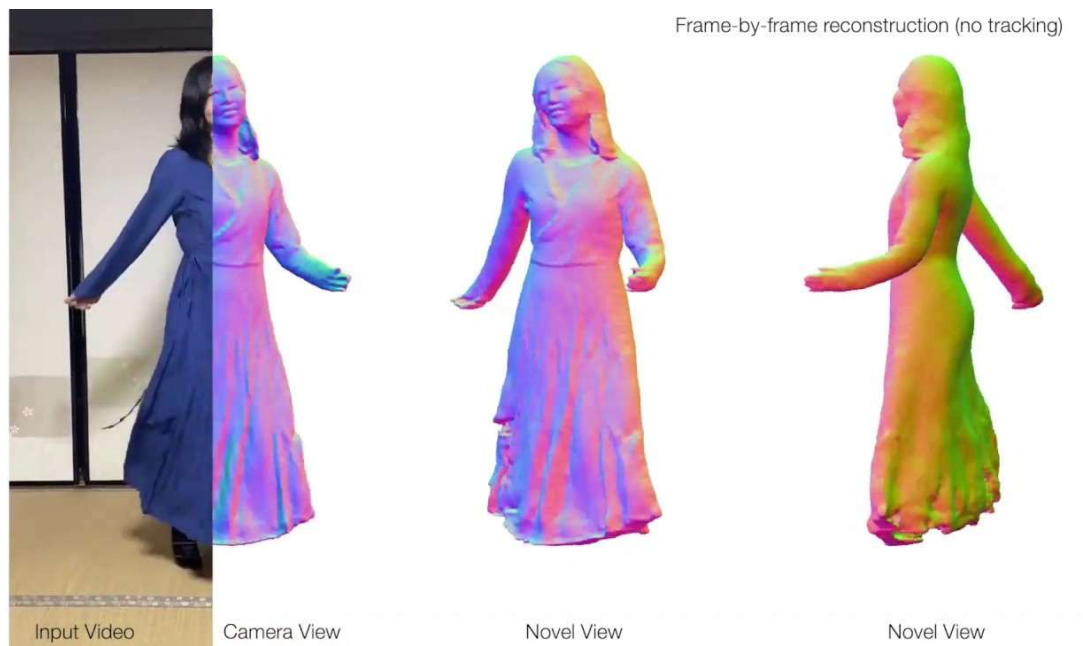


Figure 2: PIFuHD reconstruction output

7. ICON (Implicit Clothed Humans Obtained from Normals)

Overview

ICON improves clothed human reconstruction by integrating normal maps with implicit representations.

Features

- Detailed clothing reconstruction
- Improved body geometry
- Better cloth representation
- High visual realism

Advantages

- Accurate clothing reconstruction
- Enhanced body shape estimation
- Better realism than traditional methods

Limitations

- Higher computational requirements
- Complex implementation

Applications

- Virtual Fashion
- Avatar Creation
- Digital Humans
- Virtual Try-On Systems



Figure 3: ICON clothed human reconstruction

8. HumanNeRF

Overview

HumanNeRF is based on Neural Radiance Fields (NeRF) and reconstructs dynamic human avatars using neural rendering techniques.

Features

- Dynamic avatar generation
- Neural rendering

- Realistic view synthesis
- High-quality appearance modelling

Advantages

- Exceptional visual quality
- Supports motion reconstruction
- Realistic rendering

Limitations

- Very high computational cost
- Slow training process

Applications

- AR/VR
- Digital Humans
- Metaverse Platforms
- Advanced Virtual Try-On Systems

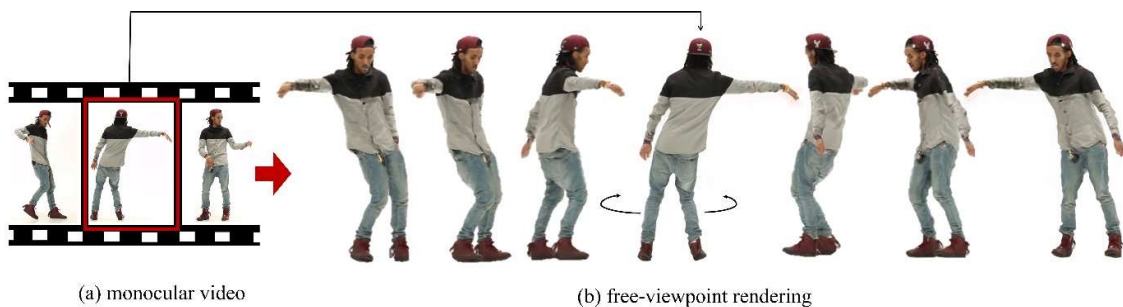


Figure 4: HumanNeRF rendering example

9. 3D Gaussian Splatting

Overview

3D Gaussian Splatting is a recent rendering approach that creates highly realistic 3D scenes and human representations using Gaussian primitives.

Features

- Real-time rendering
- High visual quality
- Efficient scene reconstruction
- Photorealistic output

Advantages

- Fast rendering speed
- Excellent visual realism
- Suitable for interactive applications

Limitations

- Emerging technology
- Limited adoption compared to traditional approaches

Applications

- AR/VR
- Real-Time Rendering
- Virtual Humans
- Interactive Fashion Systems



Figure 5: 3D Gaussian Splatting example

10. Technology Comparison

Technology	Input Type	Quality	Speed	Realism	Computational Cost
SMPL	Body Parameters	Medium	High	Medium	Low
PIFuHD	Single Image	High	Medium	High	High
ICON	Single Image	Very High	Medium	Very High	High
HumanNeRF	Multi-View Images	Very High	Low	Extremely High	Very High
3D Gaussian Splatting	Images/Video	Very High	High	Extremely High	Medium

11. Findings and Analysis

The study revealed that different technologies serve different purposes.

SMPL

Best suited for:

- Fast avatar generation
- Body measurement systems
- Virtual Try-On foundations

PIFuHD

Best suited for:

- Detailed body reconstruction
- Fashion applications
- Single-image avatar generation

ICON

Best suited for:

- Clothing-aware reconstruction
- Fashion technology
- High-quality avatars

HumanNeRF

Best suited for:

- Research environments
- High-end rendering

- Realistic digital humans

3D Gaussian Splatting

Best suited for:

- Real-time rendering
 - Interactive virtual experiences
 - Next-generation virtual humans
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12. Applications in Virtual Try-On Systems

Modern Virtual Try-On systems use 3D body reconstruction to:

- Generate user avatars
- Estimate body dimensions
- Simulate cloth behaviour
- Improve garment fitting
- Enhance realism
- Support personalized recommendations

These technologies help create immersive digital shopping experiences.

13. Challenges Identified

Several challenges remain in 3D avatar generation:

Body Shape Variations

Different body types require adaptive modelling techniques.

Clothing Complexity

Loose garments and layered clothing are difficult to reconstruct.

Computational Requirements

High-quality models require significant GPU resources.

Data Availability

Large datasets are necessary for training advanced models.

Real-Time Processing

Balancing speed and realism remain challenging.

14. Future Scope

Future developments are expected to include:

- Real-time avatar generation
 - AI-powered body reconstruction
 - Improved cloth simulation
 - Better AR/VR integration
 - Personalized digital fashion experiences
 - Advanced virtual shopping environments
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15. Learning Outcomes

During this research, the following concepts were studied:

- 3D avatar generation
 - Human body reconstruction
 - Neural rendering
 - Parametric body models
 - Digital humans
 - Computer vision workflows
 - Virtual Try-On technologies
 - AI-based human modelling
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16. Conclusion

This research explored modern techniques for 3D avatar generation and human body reconstruction, including SMPL, PIFuHD, ICON, HumanNeRF, and 3D Gaussian Splatting. Each technology offers unique advantages in terms of realism, reconstruction quality, and computational efficiency.

The findings indicate that SMPL provides a strong foundation for body modelling, while PIFuHD and ICON deliver improved reconstruction quality for fashion applications. HumanNeRF and 3D Gaussian Splatting represent the next generation of highly realistic digital human technologies. Overall, 3D avatar generation and body reconstruction play a crucial role in the future of Virtual Try-On systems, digital fashion, AR/VR experiences, and personalized e-commerce solutions.